Set 20: Stock Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse stock prices using a line chart of three stocks, a bar chart of daily percentage change for Stock A, and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
stocks <- data.frame(
  Date    = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)
write.csv(stocks, "Stock_Prices.csv", row.names = FALSE)  # for Tableau
long <- data.frame(Date = rep(stocks$Date, 3),
                   Stock = rep(c("Stock A", "Stock B", "Stock C"), each = 3),
                   Price = c(stocks$Stock_A, stocks$Stock_B, stocks$Stock_C))
pal <- c("Stock A" = "#0072B2", "Stock B" = "#E69F00", "Stock C" = "#009E73")
```

Q1. Line Chart – Stock Prices of A, B and C

> *Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.*

R Program:

```
print(stocks, row.names = FALSE)
cat("Change over the period:\n")
```

```
print(data.frame(Stock = c("A", "B", "C"),
                 Change = c(diff(range(stocks$Stock_A)), stocks$Stock_B[3] -
stocks$Stock_B[1], stocks$Stock_C[3] - stocks$Stock_C[1])))
p_line <- ggplot(long, aes(x = Date, y = Price, colour = Stock)) +
  geom_line(linewidth = 1.1) + geom_point(size = 3.5) +
  geom_text(aes(label = Price), vjust = -1.1, size = 3.4, show.legend = FALSE) +
  scale_colour_manual(values = pal) +
  scale_x_date(date_labels = "%d %b %Y", breaks = stocks$Date, expand = expansion(mult = 0.1))
+
  scale_y_continuous(limits = c(90, 165)) +
  labs(title = "Stock Prices of A, B and C (1-3 Jan 2023)", x = "Date", y = "Price", colour =
"Stock") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_line)
ggsave("s20_q1.png", p_line, width = 7.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
       Date Stock_A Stock_B Stock_C
 2023-01-01     100     150     120
 2023-01-02     105     152     118
 2023-01-03     110     148     122
Change over the period:
  Stock Change
1     A     10
2     B     -2
3     C      2
```

Output:

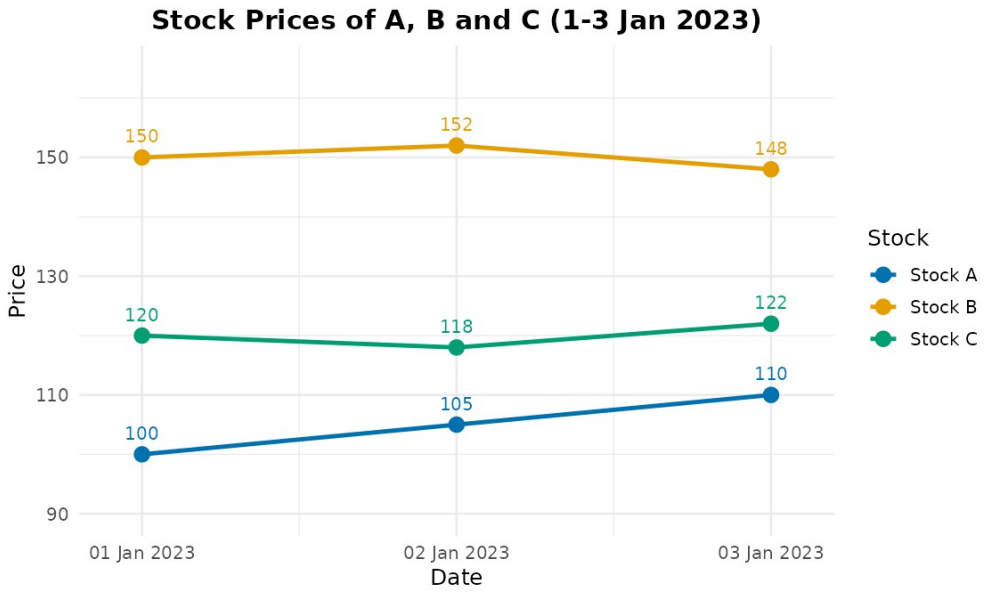


Figure 1: Stock prices from 1 to 3 January 2023

Inference:

Stock A rose steadily from 100 to 110 (+10%). Stock B is the highest-priced but fluctuated (150 → 152 → 148, a net fall of 2). Stock C dipped and recovered (120 → 118 → 122, a net rise of 2). Stock A is the best performer over the period.

Q2. Bar Chart – Daily Percentage Change of Stock A

> *Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.*

R Program:

```
chg <- data.frame(Date = stocks$Date[-1],
                  Pct_Change = round(100 * diff(stocks$Stock_A) / head(stocks$Stock_A, -1),
2))
print(chg, row.names = FALSE)
chg$Day <- factor(format(chg$Date, "%d %b %Y"), levels = format(chg$Date, "%d %b %Y"))
p_bar <- ggplot(chg, aes(x = Day, y = Pct_Change)) +
  geom_col(fill = "#0072B2", width = 0.5) +
  geom_text(aes(label = paste0("+", Pct_Change, "%")), vjust = -0.5, size = 4.5) +
  scale_y_continuous(limits = c(0, 6), labels = function(x) paste0(x, "%")) +
  labs(title = "Daily Percentage Change in Stock A", x = "Date", y = "Change from Previous Day
(%)") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_bar)
ggsave("s20_q2.png", p_bar, width = 6.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
      Date Pct_Change
 2023-01-02       5.00
 2023-01-03       4.76
```

Output:

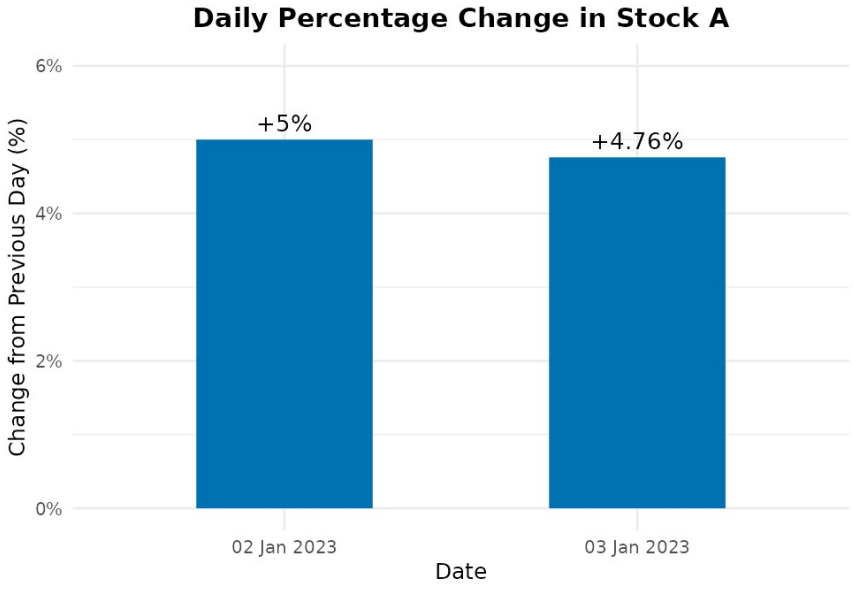


Figure 2: Daily percentage change in Stock A

Inference:

Percentage change = (today − yesterday) ÷ yesterday × 100. Stock A rose 5.00% on 2 Jan (100 → 105) and 4.76% on 3 Jan (105 → 110). 1 Jan has no previous day, so it has no change value. Gains continued, but the daily rate slowed slightly.

Q3. Table – Stock Price Data

> *Build a table to display the stock price data for each company over the given period. Label the table elements.*

R Program:

```
tbl <- stocks
tbl$Date <- format(tbl$Date, "%Y-%m-%d")
names(tbl) <- c("Date", "Stock A", "Stock B", "Stock C")
```

```
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = tbl_theme)
png("s20_q3.png", width = 900, height = 260, res = 150)
grid.arrange(grid::textGrob("Stock Price Data", gp = grid::gpar(fontsize = 14, fontface =
"bold")),
             tg, ncol = 1, heights = c(0.3, 1))
invisible(dev.off())
```

Console Output:

```
      Date Stock A Stock B Stock C
 2023-01-01     100     150     120
 2023-01-02     105     152     118
 2023-01-03     110     148     122
```

Table Output:

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Inference:

The table lists the daily price of each stock. On every day Stock B has the highest price and Stock A the lowest.

Q4. Tableau Dashboard – Interactive Exploration

> *Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.*

R Program:

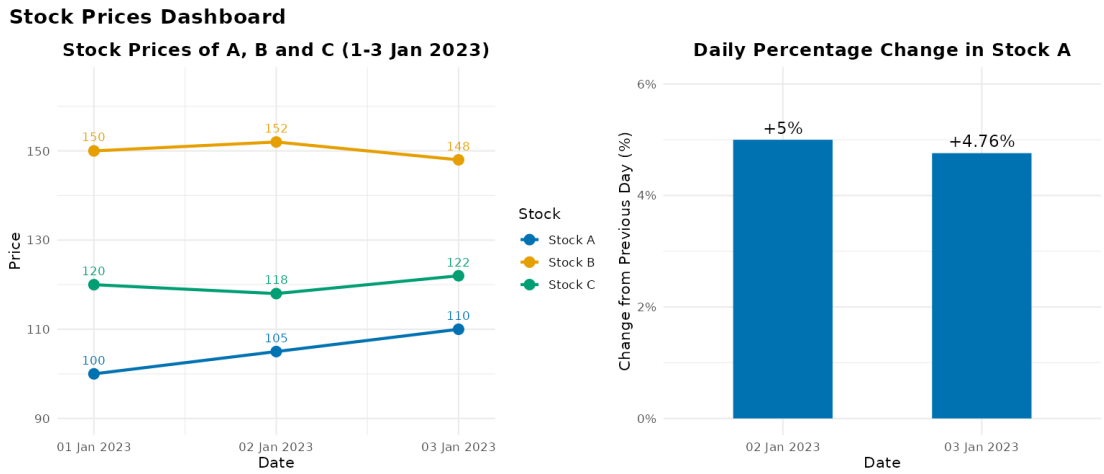
```
dash <- (p_line | p_bar) / wrap_elements(full = tg) + plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Stock Prices Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s20_dash.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Stock_Prices.csv**. Make sure Date is typed as **Date**.
2. **Sheet 1 – "Price Trend":** select Stock_A, Stock_B and Stock_C in the Data pane → right-click → **Pivot** (in the Data Source tab), and rename the fields to **Stock** and **Price**. Drag Date (Exact Date) to Columns, SUM(Price) to Rows and Stock to Color. Set the Marks type to Line with markers and labels.
3. **Sheet 2 – "Stock A % Change":** drag Date to Columns and SUM(Stock_A) to Rows. Right-click the pill → **Quick Table Calculation → Percent Difference**. Change the mark type to **Bar** and show labels (the first day is blank).
4. **Sheet 3 – "Price Table":** drag Date to Rows, Stock to Columns and SUM(Price) to Text. Rename the headers.
5. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
6. Drag the sheets onto the canvas: Price Trend (top-left), Stock A % Change (top-right), Price Table (bottom). Add a **Text** object at the top as the dashboard title.

7. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
8. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.
9. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (Stock_Prices) so one filter controls every view.
10. Add a **Stock** quick filter so users can show or hide individual stocks.
11. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Using the Stock filter isolates one company's line, and selecting a date in the bar chart highlights that day in the line chart and table.

Result

The stock prices were visualised with a line chart, a bar chart of Stock A's daily percentage change and a labelled table in R, and combined into an interactive Tableau dashboard. Stock A gained 10% over the three days (+5.00%, then +4.76%), while Stock B and Stock C moved only slightly.